

AP2: Clustering Galaxies

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Overview

This project performs density-based clustering on a dataset of 292 galaxies obtained from NASA. Each galaxy has three attributes: number of stars s_i , total brightness b_i , and density ρ_i . For clustering, only the 2D features (s_i, b_i) are used; density is included in the dataset but not part of the clustering model. All algorithms were implemented from scratch: DBSCAN and OPTICS, then connected for hierarchical clustering.

The project also compares results with built-in implementations (GeeksforGeeks / scikit-learn) to evaluate differences in clustering behavior.

Galaxies are represented as 2D points:

$$G = \{(s_1, b_1), (s_2, b_2), \dots, (s_{292}, b_{292})\}.$$

Distance metric: Euclidean distance, manually implemented:

$$d((s_i, b_i), (s_j, b_j)) = \sqrt{(s_i - s_j)^2 + (b_i - b_j)^2}.$$

DBSCAN (from scratch): identifies core galaxies with sufficient neighbors:

$$\text{Neighbors}(g_i) = \{g_j \in G : d(g_i, g_j) \leq \epsilon\},$$

$$\text{Core condition: } |\text{Neighbors}(g_i)| \geq \text{minPoints}.$$

Clusters are grown recursively from core galaxies; non-core galaxies outside dense regions are marked as noise.

OPTICS (from scratch): refines DBSCAN clusters into hierarchical subclusters. Core distance of a galaxy:

$$c(g_i) = \text{distance to } (\text{minSamples} - 1)^{\text{th}} \text{ nearest neighbor}.$$

Reachability distance:

$$r(g_j) = \max(c(g_i), d(g_i, g_j)).$$

OPTICS orders galaxies by reachability distance and extracts subclusters using a threshold, revealing internal cluster structure.

Approach

- Load the galaxy dataset into a numpy array.
- Compute Euclidean distances manually for all galaxy pairs.
- Implement DBSCAN from scratch:
 - Identify core galaxies and recursively expand clusters.
 - Assign noise to isolated galaxies.
- Implement OPTICS from scratch:
 - Compute core distances and reachability distances.
 - Order galaxies and extract hierarchical subclusters.
- Connect DBSCAN and OPTICS:
 - Run OPTICS on each DBSCAN cluster with at least minPoints.
 - Visualize subclusters with different markers while preserving DBSCAN cluster colors.
- Compare with built-in implementations (GeeksforGeeks / scikit-learn):
 - Run DBSCAN and OPTICS using library functions.
 - Observe differences in clustering due to handling of edge cases and noise.
- Test with different datasets to validate robustness and visualize clustering variations.

Experiment

DBSCAN parameters: $\epsilon = 5$, minPoints = 3. OPTICS parameters: minSamples = 3, reachability threshold computed as 70th percentile of finite distances.

Different datasets were tested to evaluate how the algorithm handles clusters of various shapes and densities. Both the from-scratch and built-in implementations were compared.

Observations

- DBSCAN effectively identifies dense galaxy clusters and isolates sparse galaxies as noise.
- OPTICS reveals hierarchical subclusters within each DBSCAN cluster, highlighting finer structures.
- Manual implementation provides insight into the mechanics of clustering, but built-in versions sometimes produce slightly different clusters due to handling of edge cases and internal optimizations.
- Testing with multiple datasets showed that small clusters or close edge points can be treated differently in the from-scratch versus library implementations.
- Visualization using same-color DBSCAN clusters and different OPTICS markers makes hierarchical structure visually clear.

Conclusions

- Both DBSCAN and OPTICS were implemented from scratch and successfully connected to provide hierarchical clustering.
- From-scratch implementations allow complete control over distance calculations, cluster expansion, and reachability distances.

- Comparison with built-in implementations highlights differences due to edge-case handling and optimizations in libraries like scikit-learn.
- Testing with multiple datasets confirms the method's robustness and ability to reveal both coarse and fine-grained clustering patterns.
- The combined approach is fully reproducible and can be applied to any 2D galaxy dataset to reveal hierarchical structures.